

JAYMZ COLLINS.

DOB – 29/06/1992.

Email: Jaymz.collins@hotmail.co.uk

Tel: 07859946829

A bioinformatics graduate with a strong foundation in computational biology, statistics, and genomics, developed through an MSc in Bioinformatics and an undergraduate degree in Neuroscience. Experienced in designing and executing full-scale sequencing analysis pipelines, from raw data processing through to biological interpretation, with particular interest in neurodegenerative disease research and drug discovery. Seeking to apply strong analytical and problem-solving skills to a role in bioinformatics or computational biology.

Education.

Cardiff University.

2021 – 2024.

Neuroscience BSc - First Class Honours.

Cardiff University Academic Scholarship. Elective modules in molecular biology, genetics and genomics with additional coursework in data analysis and statistics.

Cardiff University.

2025 – 2026.

Bioinformatics MSc.

Dissertation: Computational reanalysis of single-nucleus RNA-seq data investigating Cas9D10A nickase-mediated CAG repeat contraction in a Huntington's disease mouse model.

Supervisor: Professor Vincent Dion, UK Dementia Research Institute.

Technical Skills.

Languages/Tools: Experienced in writing and maintaining scripts for R, Python, Unix/Bash, Nextflow, SLURM.

High Performance Computing (HPC): Experience running large-scale pipelines on a HPC cluster (Falcon via South Wales Supercomputing).

RNA-seq Analysis: Experience with both single cell and single nucleus RNA-seq with packages including: Seurat, Harmony integration, SCTransform, Cell Ranger, Scrublet/DoubletFinder, UMAP/clustering workflows

Next-Generation Sequencing (NGS): Raw FASTQ data processing and quality control, checksum verification, Cell Ranger alignment, sequence read alignment (STAR), read trimming (Trimmomatic), BAM file processing and duplicate marking (Picard).

Statistics: Applied statistical modelling for high-dimensional biological data, including dimensionality reduction (PCA), differential expression testing, supervised classification (XGBoost, ensemble classifiers, conditional inference trees) and unsupervised clustering (Seurat/Harmony-based clustering, k-means).

Key Projects.

MSc Dissertation — Computational Reanalysis of Single-Nucleus RNA-Seq Data in a Huntington's Disease Model | Cardiff University / UK Dementia Research Institute | Supervisor: Professor Vincent Dion

- Independent bioinformatic reanalysis of published snRNA-seq data (BioProject PRJNA1077893) investigating Cas9D10A nickase-mediated CAG repeat contraction in the R6/1 Huntington's disease mouse model.
- Processed 700GB+ of raw FASTQ data across 12 samples (2x2 design, n=3/group), including custom GRCm39 reference genome construction incorporating three transgene sequences.
- Built a complete Seurat pipeline (QC, SCTransform, Harmony integration, PCA/UMAP, clustering) on university HPC infrastructure.
- Diverged from the original study's fixed QC threshold in favour of an adaptive MAD-based filtering approach, independently justified through comparative analysis.
- Manually annotated 25 clusters with systematic contamination flagging across ~900 cells
- Identified T and B lymphocyte populations absent from the original published annotation, extending the study's cell-type resolution.

Other Experience.

Financial Administrator. *Fatra*, Cardiff.

October 2024 – March 2025.

Net Zero Energy Consultant. *Amber*, Cardiff.

June 2023 – October 2023.

Customer service advisor. *British Gas*, Cardiff.

Jan 2014 – Feb 2020.

